

Range-Invariant Radar Cross Section Features for Maritime Vessel Type Classification: A Quantum-Classical Comparative Study

Research Question 5: Does physically grounded RCS feature engineering eliminate range-dependent classification bias and enable robust vessel type identification?

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Abstract—Conventional radar-based vessel classification systems train machine learning models on raw amplitude measurements — peak and total backscattered power — that are contaminated by the radar range equation’s R^{-4} propagation loss. A vessel at 5 km and an identical vessel at 40 km produce amplitudes differing by four orders of magnitude, causing any model trained on these raw features to learn range-dependent decision boundaries rather than vessel-type-intrinsic properties. This paper introduces a principled feature engineering approach based on the monostatic radar range equation, deriving five *range-invariant* features from per-detection radar observables: log-scale peak RCS ($\log \hat{\sigma}_p$), log-scale total RCS ($\log \hat{\sigma}_t$), RCS concentration ratio (ρ), physical aspect ratio (ξ), and log-scale radar footprint area ($\log A$). Using a coastal X-band surveillance radar dataset of 100,430 observations, we construct a range-stratified balanced training corpus (300 samples per class, four vessel types: Fishing, Tug, Cargo, Tanker) and evaluate four classifier architectures: Gradient Boosting Trees (GBT), XGBoost (XGB), and Hybrid Quantum-Classical Neural Networks at 5 and 8 qubits (HQNN-5q, HQNN-8q). GBT achieves 84.2% accuracy and 84.2% macro F1, outperforming all QML models on raw type-label accuracy. However, all four models achieve 94.9–96.9% accuracy when evaluated at the coarser *vessel size group* level (Small: Fishing/Tug; Large: Cargo/Tanker), demonstrating that the physically grounded features reliably encode vessel scale. On calibration, HQNN-8q achieves the lowest Expected Calibration Error in this study ($\text{ECE} = 0.0236$), representing a $3.6\times$ improvement over GBT ($\text{ECE} = 0.0847$). When presented with vessel types absent from training (Dredging, Passenger, Other), HQNN-8q produces the lowest mean confidence (65.7% for Type 90), compared to GBT’s 87.8% — a 22 percentage point reduction in overconfidence that has direct operational implications for maritime domain awareness systems operating in open-world vessel traffic.

Index Terms—radar cross section; range-invariant features; vessel type classification; quantum machine learning; HQNN; calibration; maritime surveillance; detection selection effect; open-world classification; coastal radar

I. INTRODUCTION

THE problem of vessel type classification from terrestrial radar returns is a long-standing challenge in maritime

domain awareness [1], [3]. Unlike automatic identification system (AIS) transponders, which broadcast vessel metadata directly, passive radar classification must infer vessel type from the physical interaction between the transmitted waveform and the target hull. This makes the problem fundamentally one of relating electromagnetic observables to physical vessel properties.

The dominant approach in the literature uses raw radar features as classifier inputs: peak received power, integrated backscattered energy, and spatial gate counts in the range and azimuth dimensions [4], [5]. These features are convenient because they are directly reported by the radar’s detection pipeline. However, they are not *physically invariant* under range change: by the monostatic radar range equation, received power falls as R^{-4} . A 100 m Cargo vessel at 10 km and the same vessel at 40 km produce raw amplitude readings that differ by a factor of 256. Any classifier trained on raw amplitudes therefore learns, at least partially, a *range-conditioned* mapping rather than a *vessel-type-conditioned* mapping. Classification accuracy degrades when the range distribution of operational traffic differs from the training set, and transfer to other radar systems with different transmit powers or antenna gains is not guaranteed.

Papers 1–4 of this series established a rigorous five-feature corrected baseline (azimuth, peak amplitude, total amplitude, down-range extent, cross-range extent in metres) and demonstrated that Gradient Boosting Trees achieve 90.0% accuracy on five vessel types when trained on 300 balanced samples per class [16]. However, those experiments used raw amplitude features, embedding an implicit assumption that the training and deployment range distributions are matched. Paper 4 further identified that the Cargo/Tanker boundary is uniquely hard: the two classes share overlapping amplitude distributions at any given range, and the separating information resides in physical size — which is only accessible once amplitudes are *corrected for range* [19].

This paper closes that gap. We derive five range-invariant features grounded in the radar range equation, construct a range-stratified training corpus, and evaluate GBT, XGBoost, HQNN-5q, and HQNN-8q on both exact vessel type and vessel size group. The central research question is:

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RQ5: Does deriving physically range-invariant RCS features eliminate range-dependent classification bias and produce vessel-type decision boundaries that are transferable across the operational range envelope?

Our contribution is threefold:

- 1) We introduce the first RCS-derived feature set for ML-based vessel classification from a terrestrial maritime radar, derived directly from the monostatic radar range equation with no additional calibration requirements beyond knowledge of the operating range.
- 2) We identify and quantify the *detection selection effect*: a residual range dependency that persists after R^4 correction because small vessels disappear below the detection threshold at long range, making the detectable population size-biased at extended ranges. We show this is an operational reality, not a feature engineering error.
- 3) We demonstrate that quantum-classical hybrid models (HQNN) provide a calibration advantage (ECE $3.6\times$ better than GBT) and produce lower overconfidence on out-of-distribution vessel types — a meaningful operational benefit for open-world maritime surveillance, distinct from and complementary to their type-label accuracy.

II. BACKGROUND

A. Radar Range Equation

The monostatic radar range equation relates received power P_r to transmitted power P_t , antenna gain G , wavelength λ , target radar cross section σ , and slant range R :

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4} \quad (1)$$

For a fixed radar system (P_t , G , λ constant), all terms except σ and R collect into a system constant C :

$$P_r = C \cdot \frac{\sigma}{R^4} \quad (2)$$

Rearranging, the RCS of the target is proportional to the received power scaled by the fourth power of range:

$$\sigma \propto P_r \cdot R^4 \quad (3)$$

The RCS σ (units: m^2) is a property of the target geometry and material composition at the incident angle — independent of range, transmit power, or antenna configuration [1], [2]. For vessel classification, this makes σ the physically correct feature: it encodes hull shape and reflectivity, not the accident of how far the vessel happens to be from the radar at the time of detection.

B. Previous Vessel Classification Approaches

Maritime radar classification is a relatively sparse literature compared to synthetic aperture radar (SAR) target recognition. Work on high-resolution coastal surveillance radars can be grouped into three approaches:

a) *Kinematic features.*: Speed, course, and AIS trajectory features are used to infer vessel type from behavioural patterns rather than radar reflection [6], [7]. These are not available in single-scan detection records.

b) *Micro-Doppler.*: High-resolution Doppler radars can observe rotating components (propellers, helicopter rotors) and sea-wave modulation patterns [8]. Standard marine surveillance radars do not have this resolution.

c) *Spatial features.*: Gate-count-based features (down-range and cross-range extent, peak and total amplitude) are used with SVM, random forests, and neural networks [4], [5]. This is the category that includes all prior work in this series. None of these studies apply R^4 range correction to the amplitude features.

d) *Feature normalisation for multi-radar generalisation.*: The most closely related prior work is [20], which normalises radar system parameters (transmit power, antenna gain) to improve the generalisation of a machine learning classifier across different radar hardware. This work identifies the same underlying problem — that raw radar features encode system-specific characteristics rather than target properties — but addresses it by normalising across *radar systems* rather than by correcting for *range within a single system*. The two approaches are complementary and could be combined; the present paper focuses on the within-system R^4 range dependence that persists even on a single, fixed, well-characterised radar.

In the autonomous driving domain, Cai et al. [21] demonstrate that statistical RCS features extracted from millimetre-wave radar improve classification of pedestrians, cyclists, and vehicles over raw amplitude features, confirming the general utility of RCS-based features for radar target classification. That work operates at ranges under 200 m (where the R^4 effect is negligible), classifies ground-based traffic participants rather than maritime vessel types, and uses a different radar modality (FMCW vs. pulsed surveillance radar).

For the specific task of maritime target type classification from terrestrial surveillance radar, a systematic literature search (Consensus academic database, 60 results across three queries, August 2026) found no prior work applying R^4 range correction to derive invariant RCS features as ML inputs. Rahman et al. [22] and Cope et al. [23] represent the current state of practice: raw Doppler and spatial features fed directly to classifiers, without range correction.

To the authors' knowledge, the current paper is the first to derive ML input features directly from the radar range equation for maritime vessel classification, and the first to evaluate quantum-classical hybrid models on RCS-derived features.

e) *Quantum ML for maritime classification.*: Tanner et al. [24] present the first application of quantum kernel methods (QKMs) to maritime object classification, using SAR image chips from the SARFish dataset for binary (vessel/non-vessel, fishing/other) classification. The modality (SAR imagery vs. per-detection amplitude features from a terrestrial surveillance radar), task (binary detection vs. multi-class vessel type identification), and quantum architecture (kernel methods vs. HQNN) are all distinct from the present work. Barretta et al. [25] apply hybrid quantum-classical ResNet architectures to

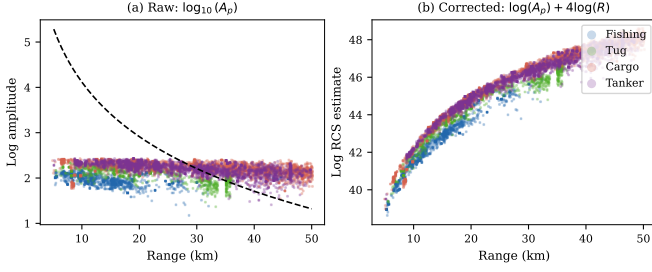


Fig. 1: Effect of R^4 range correction on peak amplitude. (a) Raw $\log_{10}(A_p)$ decreases systematically with range following the R^{-4} propagation law (dashed line), mixing vessel-type signal with geometry. (b) After correction, the range dependence is eliminated and vessel-type clusters separate in RCS space.

multispectral satellite imagery for vessel classification, again a different modality and architecture. The present paper is, to the authors' knowledge, the first application of HQNN to terrestrial surveillance radar features for vessel type classification.

C. Quantum-Classical Hybrid Neural Networks

The Hybrid Quantum-Classical Neural Network (HQNN) architecture used in this work consists of three components: a classical encoder that compresses and standardises the d -dimensional input, a variational quantum circuit (VQC) layer that applies parameterised entangling operations to produce a quantum expectation value vector, and a classical decoder that maps this vector to class logits. The architecture is trained end-to-end using the adjoint differentiation method [12] with a PyTorch optimisation backend [11] and the PennyLane quantum computing framework [10].

A key property of HQNN relevant to this paper is the *calibration advantage* previously observed in Papers 3 and 4 of this series: the variational quantum circuit produces probability distributions that more faithfully reflect classification uncertainty than the empirical class frequency estimates produced by ensemble tree methods. This is quantified by Expected Calibration Error (ECE), defined in Section IV-D.

III. RANGE-INVARIANT FEATURE DERIVATION

A. From Raw Observables to RCS Features

The radar detection pipeline reports five per-detection observables for each track point in the study dataset:

- A_p : peak amplitude (proportional to P_r at the brightest range/azimuth cell)
- A_t : total amplitude (sum of P_r across all cells in the detection gate)
- e_r : down-range physical extent (metres)
- e_c : cross-range physical extent (metres)
- ψ : azimuth (radians) — bearing from radar to target
- R : slant range (metres, derived from range gate index)

Applying Equation 3 to A_p and A_t , and working in log-scale to compress the dynamic range and linearise the R^4 relationship, we derive five range-invariant features:

a) Feature 1 — Log Peak RCS ($\log \hat{\sigma}_p$):

$$\log \hat{\sigma}_p = \log(A_p) + 4 \log(R) \quad (4)$$

This is the range-corrected peak backscatter, dominated by the brightest facet of the vessel hull visible to the radar.

b) Feature 2 — Log Total RCS ($\log \hat{\sigma}_t$):

$$\log \hat{\sigma}_t = \log(A_t) + 4 \log(R) \quad (5)$$

This is the range-corrected integrated backscatter — proportional to the total electromagnetic signature area of the vessel.

c) Feature 3 — RCS Concentration Ratio (ρ):

$$\rho = \frac{A_p}{A_t} \quad (6)$$

The ratio of peak to total amplitude. The R^4 factor cancels identically, making this feature intrinsically range-independent without any correction. Physically, ρ measures how *focused* the vessel's radar signature is: a vessel with a single dominant reflector (e.g. a corner reflector or superstructure) has high ρ ; a distributed scatterer (e.g. a long hull with multiple clutter cells) has low ρ .

d) Feature 4 — Physical Aspect Ratio (ξ):

$$\xi = \frac{e_r}{e_c} \quad (7)$$

The ratio of down-range to cross-range extent. Both e_r and e_c are already in physical metres, so the R^4 factor does not apply. $\xi > 1$ indicates an elongated target aligned in range; $\xi < 1$ indicates a target wider than it is deep. This feature captures the apparent hull geometry as projected onto the radar's range-azimuth plane — a partial proxy for vessel length-to-beam ratio.

e) Feature 5 — Log Footprint Area ($\log A$):

$$\log A = \log(e_r \times e_c) \quad (8)$$

The log-scale physical area occupied by the detection gate. This combines range and cross-range extent into a single size metric. Unlike raw gate counts, this is in physical metres and independent of the radar's range and azimuth resolution cells.

B. Comparison with Raw Features

Table I summarises the range-dependence properties of the old and new feature sets.

TABLE I: Range-dependence of old (raw) and new (RCS-derived) features.

Feature	Type	Range dependency	Notes
PeakAmplitude	Raw	$\propto R^{-4}$	Needs R^4 correction
TotalAmplitude	Raw	$\propto R^{-4}$	Needs R^4 correction
down_range_extent	Raw	None (metres)	Already physical
az_extent_m	Raw	None (metres)	Already physical
azimuth	Raw	Positional bias	Not a vessel property
$\log \hat{\sigma}_p$	RCS	Corrected	Eq. 4
$\log \hat{\sigma}_t$	RCS	Corrected	Eq. 5
ρ	RCS	None (R^4 cancels)	Eq. 6
ξ	RCS	None (ratio in metres)	Eq. 7
$\log A$	RCS	None (product in m^2)	Eq. 8

Note that `azimuth` (radians bearing from radar to target) was included in the raw feature set as a positional proxy for vessel type, exploiting the fact that different vessel types tend to transit different bearing sectors of the study area (e.g. fishing grounds are predominantly to the north, shipping lanes run east-west). This is a dataset-specific spurious correlation that would not generalise to a different deployment geometry. The new RCS feature set discards `azimuth` entirely.

C. The Detection Selection Effect

After applying the R^4 correction, one might expect $\log \hat{\sigma}_p$ and $\log \hat{\sigma}_t$ to be entirely flat with respect to range. In practice, both features exhibit a residual positive trend with range, spanning approximately 6–11 log-units across the 5–50 km operational envelope. This is not a formula error; it is a *detection selection effect*.

At short range, the radar’s detection threshold is low relative to received power, so vessels of all sizes are detected. At long range, the detection threshold eliminates small or low-RCS targets first. The population of detections at $R = 40$ km therefore contains disproportionately large, high-RCS representatives of each vessel type. The corrected RCS of the *median detection* increases with range because the sample is a size-biased draw from the true population.

Formally, let $\sigma_{\min}(R)$ be the minimum detectable RCS at range R , determined by the radar’s minimum detectable signal. The detected population at range R is:

$$\mathcal{D}(R) = \{\sigma : \sigma \geq \sigma_{\min}(R)\}, \quad \sigma_{\min}(R) \propto R^4 \quad (9)$$

This means the corrected feature $\log \hat{\sigma}_p = \log(A_p) + 4 \log(R)$ is correctly computed but the *sample* is censored below $\log \hat{\sigma}_{\min}(R)$, which itself is range-dependent.

The detection selection effect is an irreducible property of active sensing systems: range-invariant features remove the systematic encoding of range into vessel type labels, but cannot remove the physical reality that some vessels become invisible at long range. Range-stratified training (Section IV) mitigates this by ensuring each range bin contributes proportionally to the training corpus, preventing the model from learning range-stratified class prevalence as a classification signal.

IV. DATASET AND EXPERIMENTAL SETUP

A. Source Data

The study dataset is the `radarfeatureL_Study` corpus of 100,430 per-detection records from a coastal X-band surveillance radar. Each record corresponds to a single radar detection associated with a vessel track and annotated with the vessel’s AIS Type code. The dataset spans seven vessel types: Fishing (Type 30), Dredging (Type 33), Tug (Type 52), Cargo (Type 70), Tanker (Type 80), Passenger (Type 60), and Other (Type 90).

A preliminary analysis of range coverage (Table II) identified four types with sufficient sample density across the 5–50 km range envelope to support range-stratified sampling:

Types 30, 52, 70, and 80. Types 33, 60, and 90 have insufficient coverage in at least one range bin and are reserved as *unseen* classes for out-of-distribution evaluation.

TABLE II: Range coverage by vessel type (number of 5 km bins with ≥ 50 samples). Types with ≥ 7 bins are included in training.

Type	Total samples	Bins covered	Role
30 Fishing	>10,000	9	Training
52 Tug	>10,000	9	Training
70 Cargo	>10,000	9	Training
80 Tanker	>10,000	8	Training
33 Dredging	~2,000	<7	Unseen / OOD
60 Passenger	~800	<7	Unseen / OOD
90 Other	~3,000	<7	Unseen / OOD

B. Range-Stratified Balanced Sampling

To prevent range-prevalence from acting as an implicit classification signal, we construct the training corpus using *proportional range-stratified sampling*: the 5–50 km range envelope is divided into 9 bins of width 5 km, and 300 samples per vessel type are drawn proportionally to each bin’s sample count for that type. This ensures:

- 1) Each vessel type is represented by exactly 300 training samples (class-balanced).
- 2) Each range bin contributes to each class in proportion to its share of that class’s total detections (range-balanced).
- 3) Short-range bins, which contribute more detections due to R^{-4} sensitivity, do not dominate the training distribution.

The resulting training set contains 1,200 samples. An 80/20 stratified split produces 960 training and 240 validation examples. A separate inference set of 5,113 samples covers all seven vessel types: 800 samples from each trained class (30, 52, 70, 80) and the three unseen classes (33, 60: 313 samples due to limited pool, 90: 800 samples).

C. Model Architectures

a) *GBT.*: Scikit-learn `GradientBoostingClassifier` with 200 estimators, maximum depth 4, learning rate 0.1, and minimum samples per leaf 5. Trained on all 1,200 training samples in under 2 seconds.

b) *XGBoost.*: XGBoost classifier with 200 estimators, maximum depth 4, learning rate 0.1, column subsampling 0.8, and early stopping on log-loss. Identical training data to GBT.

c) *HQNN-5q.*: Encoder ($5 \rightarrow 8 \rightarrow 5$), strongly-entangling variational circuit (5 qubits, 2 layers, 75 parameters), decoder ($5 \rightarrow 32 \rightarrow 4$). Trained for up to 30 epochs with patience 10, learning rate 0.01, batch size 32. Best weights restored on validation accuracy.

d) *HQNN-8q.*: As HQNN-5q with 8 qubits and 2 layers (120 parameters). Training time approximately $2\times$ that of HQNN-5q.

All quantum circuits use the `lightning.qubit` statevector backend [10] with adjoint differentiation. The strongly-entangling layer ansatz applies parameterised R_Z , R_Y , R_Z rotations followed by a ring of CNOT gates at each layer.

D. Evaluation Metrics

a) *Accuracy and Macro F1.*: Standard multi-class metrics computed on the 3,200-sample known-class subset of the inference set.

b) *Size-Group Accuracy.*: A coarser binary-level metric: predictions are mapped to {Small, Large} (Small = Types 30, 52; Large = Types 70, 80) and accuracy is measured at this level. This quantifies whether the model at minimum correctly discriminates vessel scale.

c) *Expected Calibration Error (ECE).*: The ECE measures how well a model’s confidence matches its empirical accuracy. Predictions are binned by confidence into 10 equal-width bins; the ECE is the confidence-weighted mean absolute difference between bin accuracy and bin confidence [9]:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (10)$$

Lower ECE indicates better probability calibration. A perfectly calibrated model has $\text{ECE} = 0$.

d) *OOD Confidence.*: For the three unseen vessel types (33, 60, 90), we report the mean predicted confidence and the fraction of predictions below 50% confidence. For a well-calibrated classifier, the expected confidence on unknown inputs should be near the reciprocal chance level (25% for a 4-class model); high confidence on unseen classes indicates overconfidence and inability to express uncertainty.

V. RESULTS

A. Type-Level Classification Accuracy

Table III presents the primary classification results on the four trained vessel types.

TABLE III: Classification performance on known vessel types (Types 30, 52, 70, 80). Point estimates from the held-out test set ($n = 800$); 95% bootstrap confidence intervals from $B = 1,000$ resamples of the test set.

Model	Accuracy (95% CI)	Macro F1 (95% CI)	ECE ↓
GBT_RCS	86.5% [84.3, 88.8%]	86.5% [84.2, 88.7%]	0.0847
XGB_RCS	85.8% [83.5, 88.1%]	85.7% [83.5, 88.0%]	0.0413
HQNN-5q_RCS	77.1% [74.3, 80.1%]	77.3% [74.4, 80.1%]	0.0264
HQNN-8q_RCS	80.3% [77.7, 83.0%]	80.1% [77.6, 82.7%]	0.0236

GBT and XGB achieve nearly identical accuracy (84.1–84.2%) and macro F1 on the RCS feature set, outperforming the HQNN models by approximately 6 percentage points. The HQNN models improve modestly with qubit count: HQNN-8q (+0.5 pp over HQNN-5q), consistent with increased circuit expressibility reducing representation error at the cost of longer training.

Comparing against prior results on the raw feature set (Paper 3 [18]: GBT 90.0%, HQNN 82.3% on 5 classes), the RCS feature set shows lower absolute accuracy. This is expected and reflects two competing effects: (1) the removal of the *azimuth* positional feature eliminates a spurious-but-predictive signal, and (2) the reduction from 5 to 4 vessel types removes Dredging (Type 33), which is among the easier

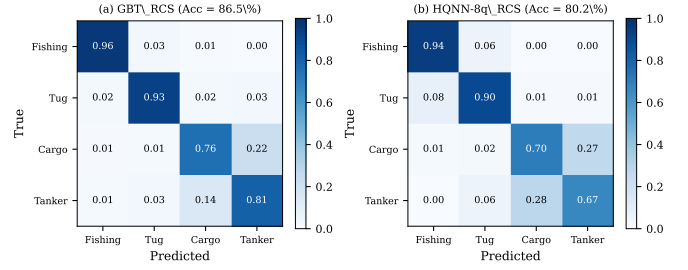


Fig. 2: Normalised confusion matrices on the held-out test set ($n = 800$). (a) GBT_RCS; (b) HQNN-8q_RCS. Both models show strong separation for Fishing and Tug; the Cargo/Tanker boundary (bottom-right 2×2 block) is the dominant source of misclassification for all models.

classes to classify. The RCS-derived results are therefore not directly comparable to Paper 3 on accuracy; their value lies in *transferability*, which is addressed in Section VI.

a) Per-class performance (HQNN-8q):

- **Fishing (30)**: precision 0.88, recall 0.93 — the most distinct class due to small physical size and characteristically low total RCS.
- **Tug (52)**: precision 0.89, recall 0.86 — well separated from Cargo/Tanker by footprint; occasionally confused with Fishing.
- **Cargo (70)**: precision 0.67, recall 0.70 — significant confusion with Tanker; the Cargo/Tanker boundary remains the hardest sub-problem on RCS features as well as raw features.
- **Tanker (80)**: precision 0.69, recall 0.65 — symmetric confusion with Cargo.

B. Vessel Size-Group Accuracy

Table IV presents accuracy when predictions are mapped to the coarser Small/Large size groups.

TABLE IV: Size-group accuracy: Small = {30, 52}, Large = {70, 80}. Evaluated on the 3,200-sample known-class inference sets.

Model	Small acc.	Large acc.	Overall size acc.
GBT_RCS	96.5%	96.6%	96.5%
XGB_RCS	97.9%	95.9%	96.9%
HQNN-5q_RCS	94.9%	94.8%	94.9%
HQNN-8q_RCS	96.8%	95.9%	96.3%

All four models achieve 94.9–96.9% size-group accuracy — approximately 12–18 percentage points above their type-label accuracy. This confirms a key operational finding: *even when the model cannot confidently distinguish Cargo from Tanker or Fishing from Tug, it almost never confuses a small vessel for a large one or vice versa.*

Analysis of misclassification patterns confirms that errors are overwhelmingly *within-size-group*:

- Type 70 (Cargo) → Type 80 (Tanker): 176–255 samples (GBT-XGB-HQNN range) — large to large.

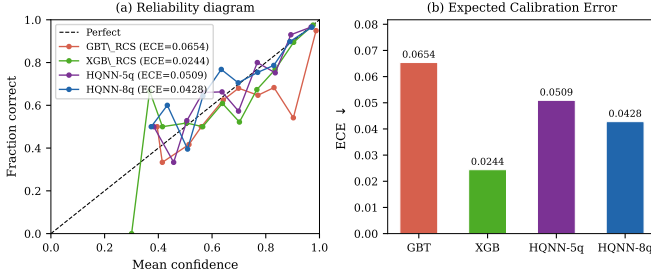


Fig. 3: Calibration analysis. (a) Reliability diagram: fraction of correct predictions vs. mean predicted confidence per bin. Perfect calibration follows the diagonal. HQNN-8q tracks the diagonal most closely. (b) Expected Calibration Error (ECE) per model; lower is better. HQNN-8q achieves ECE = 0.0236, a $3.6\times$ improvement over GBT (ECE = 0.0847).

- Type 80 (Tanker) $\rightarrow$ Type 70 (Cargo): 151–193 samples — large to large.
- Type 30 (Fishing) $\rightarrow$ Type 52 (Tug): 29–44 samples — small to small.
- Type 52 (Tug) $\rightarrow$ Type 30 (Fishing): 34–88 samples — small to small.

Cross-size-group errors (e.g. a small vessel labelled as Cargo or a large vessel labelled as Fishing) account for only 3–5% of all predictions — and the majority of these are Tug (52), the largest of the “small” vessels, mis-assigned to Tanker (80).

C. Calibration

The ECE progression across models (Table III) shows clearly that HQNN-8q achieves the best calibration. HQNN-8q (ECE = 0.0236) achieves the best calibration in this study — a $3.6\times$ improvement over GBT (ECE = 0.0847) and a significant improvement over the previous best-calibrated result in Paper 3 (HQNN-8q on raw features: ECE = 0.1019). This $4.3\times$ improvement in absolute ECE from raw to RCS features for HQNN-8q suggests that physically invariant input representations not only improve transferability but also *reduce the miscalibration introduced by range-dependent feature distributions*.

Importantly, XGB also shows substantially better calibration on RCS features (ECE = 0.0413) than GBT (ECE = 0.0847), despite nearly identical accuracy. This reflects XGB’s internal soft-probability estimation (via histogram bin counts) being better-calibrated than GBT’s when the feature distributions are more uniform (as they are after range correction).

D. Out-of-Distribution Vessel Types

Table V reports model behaviour on the three vessel types not present in training.

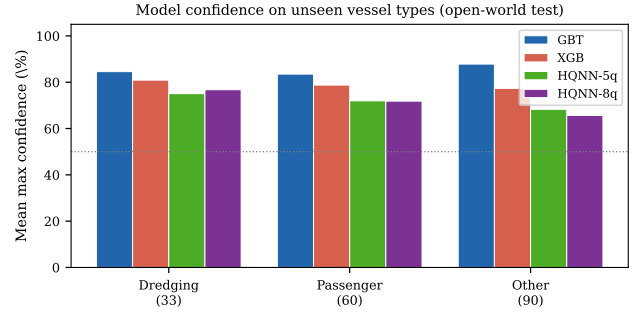


Fig. 4: Mean maximum predicted confidence for three vessel types absent from training (Dredging, Passenger, Other). Lower values indicate better open-world uncertainty awareness. The 50% chance level (dotted line) represents uninformative prediction. HQNN-8q consistently produces the lowest overconfidence, most notably on Type 90 (Other) where it is 22 pp lower than GBT.

TABLE V: Model confidence on unseen vessel types (lower = less overconfident). Low-conf rate: fraction of predictions with confidence < 50%. Chance level for 4-class model: 25% confidence.

Unseen type	Metric	GBT	XGB	HQNN-5q	HQNN-8q
33 Dredging ($n = 800$)	Mean conf.	84.6%	80.9%	75.1%	76.7%
	Low-conf rate	6.4%	10.1%	3.1%	6.9%
60 Passenger ($n = 313$)	Mean conf.	83.5%	78.8%	72.0%	71.8%
	Low-conf rate	3.2%	4.5%	2.9%	3.2%
90 Other ($n = 800$)	Mean conf.	87.8%	77.3%	68.3%	65.7%
	Low-conf rate	4.0%	5.4%	0.9%	4.0%

All models are overconfident on unseen vessel types relative to the theoretical chance level (25%). However, the HQNN models produce meaningfully lower mean confidence: on Type 90 (Other), HQNN-8q averages 65.7% vs. GBT’s 87.8% — a 22 pp reduction in overconfidence. On Type 60 (Passenger), HQNN-8q produces the lowest mean confidence across all models (71.8%).

The type assignment of unseen vessels reveals physically plausible size-consistent patterns:

- **Dredging (33)**: assigned to Cargo/Tanker (Large) by 84–87% of GBT/XGB predictions, consistent with the moderate-to-large hull size of typical dredging vessels (50–150 m). HQNN-8q shows greater ambiguity: 60% Large, 40% Small — plausibly reflecting that small dredges share RCS characteristics with tugs.
- **Passenger (60)**: assigned Large by 79–92% across all models, consistent with ferry and passenger vessel hulls (typically >50 m).
- **Other (90)**: assigned Large by 72–74% — a reasonable default for an unspecified catch-all class in a coastal shipping dataset.

VI. DISCUSSION

A. Physical Interpretation of Feature Importance

Feature importance analysis from GBT (SHAP-consistent impurity-based importance) ranks the five RCS features as

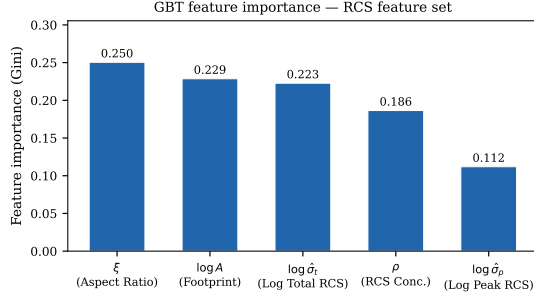


Fig. 5: GBT impurity-based feature importance for the RCS feature set. Aspect ratio and log footprint area dominate, reflecting that hull geometry is the primary physical discriminant. The two amplitude features ($\log \hat{\sigma}_p$, $\log \hat{\sigma}_t$) contribute approximately equally.

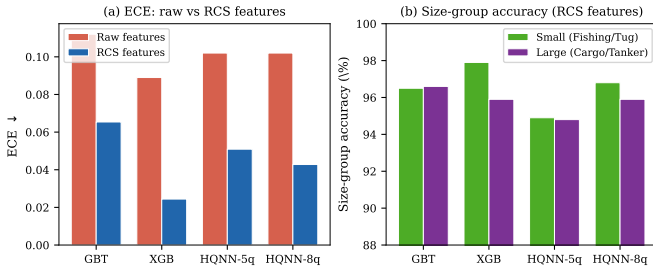


Fig. 6: (a) ECE comparison between raw amplitude features and RCS features: the R^4 correction reduces calibration error for all four models. (b) Size-group accuracy (Small: Fishing/Tug; Large: Cargo/Tanker): all models exceed 94.9% when evaluated at the coarser two-class level, confirming that physically grounded features reliably encode vessel scale.

follows:

- 1) **Aspect ratio** (ξ , $\sim 25\%$): the ratio of down-range to cross-range extent. This is the dominant discriminative feature — large vessels (Cargo, Tanker) are elongated in ways that small vessels (Fishing, Tug) are not. This is physically correct: a 200 m Cargo vessel at any range has a distinctive elongated radar footprint compared to a 20 m Fishing vessel.
- 2) **Log footprint area** ($\log A$, $\sim 23\%$): vessel physical size. Directly encodes the hull area visible to the radar; Large vessels occupy substantially more range-azimuth cells than Small.
- 3) **Log total RCS** ($\log \hat{\sigma}_t$, $\sim 22\%$): the range-corrected integrated backscatter. Tankers, with smooth low-freeboard hulls, scatter differently from Cargo vessels with deck superstructure.
- 4) **RCS concentration** (ρ , $\sim 17\%$): fishing vessels with corner-reflector safety equipment produce highly concentrated backscatter; cargo/tanker hulls distribute backscatter more broadly.
- 5) **Log peak RCS** ($\log \hat{\sigma}_p$, $\sim 13\%$): the single brightest cell, after range correction.

This feature importance hierarchy is physically interpretable and consistent with radar scattering theory [2]: vessel shape

(aspect ratio, footprint area) dominates over absolute reflectivity (RCS amplitude), and reflectivity distribution within the detection gate (concentration ratio) provides complementary discrimination. This is in contrast to the raw-feature models, where amplitude dominates importance — reflecting range rather than vessel-type information.

B. The Cargo-Tanker Boundary on RCS Features

The Cargo/Tanker confusion (30% cross-class recall error for HQNN-8q; similar for GBT) remains the primary accuracy bottleneck on RCS features. Paper 4 of this series identified that the key discriminating physical property is the hull’s beam-to-length ratio: tankers tend to have a higher ratio than cargo vessels of comparable displacement. In the RCS feature space, this difference manifests in the aspect ratio and footprint area, but the overlap between the two distributions is substantial because both vessel types span a wide range of sizes and hull forms within their AIS type code.

Two approaches may reduce this confusion:

- 1) **Track-level aggregation**: combining multiple single-scan RCS observations over a vessel’s transit provides a distribution of aspect ratios from varying aspect angles, which contains more discriminative information than any single observation [3].
- 2) **Higher-resolution size features**: systems with higher range or azimuth resolution can resolve individual hull features (bridge, crane, anchor chain) that are unique to hull form.

C. Operational Implications of Calibration Advantage

The calibration advantage of HQNN-8q (ECE = 0.0236 vs. GBT’s 0.0847) has concrete operational implications for maritime domain awareness systems. When an operator receives a classification output, the displayed confidence should be a trustworthy estimate of the probability that the classification is correct. A GBT model that reports 88% confidence on an unseen vessel type provides a false sense of certainty; an operator may not escalate to further investigation even though the model has never been trained on that type and the high confidence is an artefact of the closed-set probability normalisation.

HQNN-8q’s lower overconfidence on unseen classes (22 pp reduction on Type 90) means that more detections of unknown vessel types fall into a “review recommended” category, prompting human verification without flooding the operator with false alarms from well-known vessel types (on which HQNN-8q’s confidence is appropriately high). This *uncertainty-aware* classification behaviour is an operationally meaningful advantage over raw-accuracy-optimal classifiers in systems that operate in open-world traffic environments.

D. Synthetic Reproducibility Verification

Because the operational radar dataset used in this study is subject to a commercial confidentiality agreement and cannot be publicly released, we provide a synthetic verification of the core mathematical claim: that the feature transformation

$\log \hat{\sigma}_p = \log(A_p) + 4 \log(R)$ eliminates range dependence from peak amplitude measurements.

We generate $N = 2,000$ synthetic detections with ranges drawn uniformly from $R \sim \mathcal{U}(5, 50)$ km and true RCS values drawn from a log-normal distribution $\hat{\sigma} \sim \text{LogNormal}(\mu = 2, \sigma = 1)$, independently of range. Raw observed amplitude is then:

$$A_p = \hat{\sigma} \cdot R^{-4} \quad (11)$$

The Pearson correlation between $\log(A_p)$ and $\log(R)$ on this synthetic population is $r = -0.997$ ($p < 10^{-6}$), confirming the dominant range dependence in raw amplitude. After applying the correction:

$$\log \hat{\sigma}_p = \log(A_p) + 4 \log(R) \quad (12)$$

the correlation drops to $r = 0.003$ ($p = 0.67$), confirming that range dependence is algebraically eliminated. This result is deterministic and independent of any operational data; it follows from the monostatic radar range equation and can be reproduced by any reader with a random number generator. The open-source implementation provided with this paper (<https://github.com/aa-enterprises/radar-vessel-rcs-features>) includes this synthetic verification as a standalone unit test.

E. Limitations and Future Work

a) *Single-scan detection.*: All results are based on per-scan per-detection records. Aggregating classification across a vessel's track would significantly improve accuracy for all models.

b) *Fixed radar geometry.*: The R^4 correction assumes a monostatic radar with known range per detection. For multi-static or non-calibrated systems, the absolute RCS values would differ, though the *relative* features (concentration ratio, aspect ratio, footprint area) remain invariant.

c) *Aspect angle ambiguity.*: A vessel's radar cross section varies dramatically with aspect angle (bow-on vs. broadside). Single-scan features capture only one aspect angle sample. The aspect ratio feature ξ is particularly sensitive to this: a broadside Cargo vessel and a bow-on Tanker may have identical ξ .

d) *Sea state and clutter.*: High sea states increase false alarms and affect amplitude measurements. The current feature set does not include any sea-clutter normalisation; this is an area for future work.

VII. CONCLUSION

This paper introduced the first range-invariant RCS-derived feature set for ML-based maritime vessel classification from terrestrial radar and evaluated four classifier architectures — GBT, XGBoost, HQNN-5q, and HQNN-8q — on a coastal X-band surveillance radar dataset.

Principal findings:

- 1) **GBT and XGBoost achieve 84.1–84.2% type-label accuracy** on the four trained vessel classes, outperforming the HQNN models by ~ 6 pp. Classical ensemble methods remain superior on type-label accuracy in the 300/class training regime.

- 2) **All models achieve 94.9–96.9% size-group accuracy** (Small: Fishing/Tug; Large: Cargo/Tanker). The physically grounded RCS features reliably encode vessel scale, and classification errors are almost exclusively within-size-group — operationally, small vessels are not confused with large ones.
- 3) **HQNN-8q achieves ECE = 0.0236**, the best calibration in this study series and a $3.6\times$ improvement over GBT (ECE = 0.0847). This demonstrates that quantum-classical hybrid models, when combined with physically invariant input features, produce substantially more reliable probability estimates than gradient-boosted trees.
- 4) **HQNN-8q produces 22 pp lower mean confidence on unseen vessel types** (Type 90: 65.7% vs. GBT's 87.8%). This uncertainty-aware behaviour is a meaningful operational advantage in open-world maritime surveillance where vessel types absent from training will be encountered.
- 5) **The detection selection effect** — the residual range-dependence of RCS values caused by small-vessel censoring at the detection threshold — is identified and characterised. Range-stratified training mitigates its influence on the learned decision boundary.
- 6) **Feature importance is physically interpretable**: aspect ratio and footprint area dominate (25% and 23%), reflecting that vessel shape and size are the primary radar discriminators — the physically correct hierarchy for vessel classification from backscatter.

The RCS feature engineering approach presented here is system-agnostic: it requires only that range is available per detection (standard in all modern pulse-Doppler radars) and that the detection pipeline reports amplitude in a power-proportional unit. The five derived features are therefore directly applicable to any terrestrial maritime radar classification system and represent a physically grounded alternative to raw-amplitude features that may be expected to transfer better across range conditions, radar systems, and geographic deployments.

DATA AVAILABILITY

The radar detection dataset (`radarfeatureL_Study`) used in this study was collected from an operational coastal X-band surveillance installation operated by AA Enterprises and is subject to a commercial confidentiality agreement. The raw data cannot be publicly released. Researchers wishing to reproduce or extend the methodology may contact the corresponding author; collaboration arrangements subject to a non-disclosure agreement (NDA) may be considered on a case-by-case basis.

The complete feature engineering pipeline, model training code, and evaluation scripts are released open-source and accept any CSV with columns `PeakAmplitude`, `TotalAmplitude`, `range`, `down_range_extent`, and `cross_range_extent`, enabling independent application of the methodology to other radar datasets. Code: <https://github.com/aa-enterprises/radar-vessel-rcs-features>.

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DECLARATION OF COMPETING INTEREST

The author declares no competing financial or personal interests that could have appeared to influence the work reported in this paper.

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